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Project Title:	Performance and Sleep Consequences of Repeated Phase Shifts Within Appendix K Guidelines		
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	The aim of this study is to help understand the nature of repeated phase shifts on the human Endogenous Circadian Pacemaker (ECP), and the effects of such repeated shifts on circadian rhythms, sleep, performance, alertness and mood. This will help inform schedulers on future Space Shuttle and International Space Station missions. For any phase shift there are two ways of getting to the desired phase position: 1)Clockwise around the clock (repeated phase delays to later bedtimes), or 2) Anti-clockwise against the clock (repeated phase advances to earlier bedtimes). Thus, for example a 6h phase advance and an 18h phase delay both end up at the same phase position. Crew scheduling guidelines (Appendix K) allow repeated phase advances of up to 30 minutes and repeated phase delays of up to two hours. We hypothesize that it might be better, in some case, to accommodate a given phase advance (e.g., a 6h phase advance required for operational reasons), by giving repeated 30 minute phase advances in schedule (to earlier times), than by giving repeated 2h phase delays (to later times). The experiment will comprise a three year ground-based study of how human sleep, circadian rhythms and performance are affected by repeated phase shifts that are within NASA crew scheduling		

Task Description:	constraints. Using time isolation facilities at the University of Pittsburgh we plan to simulate two different 16-day missions. Following Appendix K guidelines and standard practice, all sleep periods (subjective "nights") will be exactly 8h in length. Both missions will start with a four-day segment which will serve as a control comparison, and will involve no phase shifts, with subjects having a 2300 bedtime and an 0700 wake time. The "delay" mission will then require repeated 2h phase delays on each of the next nine successive nights of the study (i.e., bedtimes at 0100, 0300, 0500, 0700, etc.), resulting in an eventual movement of bedtimes to an 18h phase delayed position, with a bedtime at 1700 and wake time at 0100, which will be retained for the remaining nights. The experiment will last a total of 16 calendar days (384 hours). The "advance" mission will again start with four control days, after which the protocol will require repeated 30 minute phase advances on each successive night of the study (i.e., bedtimes at 2230, 2200, 2130, 2100 etc.). This will also lead, at the completion of 12 subjective nights, to a 6h phase advanced position (a 1700 bedtime, 0100 wake time). Again, the experiment will last a total of 16 calendar days (384 hours). On all days in both missions, breakfast will be 1.5h after wake time, lunch 6h after wake time, and dinner 12h after wake time. Snacks will be available ad-lib. Showers and exercise on an exercycle (optional) will be restricted to a 1.5h period prior to lunch. Mood, alertness and performance tests will be scheduled before each meal, after lunch and before bedtime. Objective measures of sleepiness will be given after lunch. Healthy volunteers (ages 20y - 59y, both genders) will live and work in the facility singly or in groups of between two and four, and will be studied in terms of their sleep (measured polysomnographically and by diary); circadian rhythms (rectal temperature, urinary free cortisol, urinary melatonin sulfate); performance speed and accuracy (on a computerized battery of tests involving vigilance, serial reaction, response inhibition and verbal reasoning); mood, and both subjective and objective measures of alertness/sleepiness. The experiment will start at 0900 on a Monday, and end exactly 16 days later, at 0900 on a Wednesday morning.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Every day, millions of night workers and air travelers are required to change the timing of their biological clock (circadian system) so that it is correctly aligned to their new work/rest schedule. This very often causes distress related to sleep disruption and to impairments of mood, alertness and performance while awake. Such distress may in some cases lead to accidents and illness. Very often such schedule changes are difficult to make abruptly, and often it is more reasonable to adopt a more gradual approach, trickling in the time changes little by little, rather than in one acute shift in schedule. While the present task is mostly concerned with work/rest scheduling in space, it also addresses an important question for all human beings trying to phase shift their schedule, namely that of whether trickling in a change in routine by repeated phase advances or by repeated phase delays is the better way to do this. Thus earth benefits will accrue to shift workers and air travelers from this task. The knowledge about the response of humans to repeated small phase shifts derived from this research may lead to alternative strategies to ameliorate these problems.
Task Progress:	One major lesson learned early on in the manned space program was that constraints needed to be imposed upon the timing of a crew's scheduled sleep episodes. If sleep was scheduled at the "wrong" circadian time, then it was likely to be disrupted and unrefreshing. As a means of avoiding sleep disruption problems, NASA originally developed a document referred to as Appendix K which was used to specify the constraints on crew timeline scheduling. Among other goals relating to the need to avoid overburdening the crew, Appendix K sought both to ensure that enough sleep opportunity was given (8h per subjective "night"), and to ensure that the placement of the sleep did not violate certain circadian principles which related to maximum amounts by which an astronaut's bedtime could change from one day to the next. Since then several iterations of these rules have been implemented, but all work under the assumption that by "trickling in" phase shifts a little at a time, the endogenous circadian pacemaker can remain appropriately entrained (i.e. remain in correct temporal alignment to the work/sleep routine), thereby avoiding circadian dysfunction with its associated disruptions in sleep and performance functioning. These principles were based on the notion of a "directional asymmetry" whereby phase delays (to later bedtimes) were held to be less disruptive to astronauts' sleep than phase advances (to earlier bedtimes). In particular, they make the assumption that, unlike repeated equivalent phase advances, repeated phase delays of 2h per day are within the limits the endogenous circadian pacemaker and thus not likely to lead to circadian dysfunction. Thus, according to Appendix K guidelines, a 6h phase advance is better accomplished by nine repeated 2h phase delays, rather than by one abrupt 6h phase advance. PROGRESS PHASE DELAY DIRECTION Ten subjects were run in the phase delay direction (5M, 5F, mean age: 41.2y, s.d.: 13.9y). Preliminary data were analyzed and presented briefly by Dr. Monk at an oral presentation in the 2003 Bioastronautics Investigators' Workshop in Galveston, TX. We then wrote up the complete findings as a full peer-reviewed paper in Aviation, Space, and Environmental Medicine (Final title: "Using nine 2-h delays to achieve a 6-h advance disrupts sleep, alertness, and circadian rhythm" by Monk, TH, Buysse, DJ, Billy, BD and DeGrazia, JM). This paper was submitted in March 2004, revised and resubmitted in August 2004, and accepted for publication in September 2004. PHASE ADVANCE DIRECTION Eleven subjects were run in the phase advance direction. The data from one subject had to be discarded as he had failed to adhere to the 2300-0700 sleep time in the pre-laboratory baseline period at home. Data analysis on the remaining 10 subjects (7M, 3F) is ongoing. We plan to present preliminary results as a poster at the 2005 Bioastronautics Investigators' Workshop in Galveston, TX, and to write up these results for another peer-reviewed paper in Aviation, Space, and Environmental Medicine which we hope to submit in early 2005.
Bibliography Type:	Description: (Last Updated: 04/27/2011)

Articles in Peer-reviewed Journals	Monk TH, Buysse DJ, Billy BD, DeGrazia JM. "Using nine 2-h delays to achieve a 6-h advance disrupts sleep, alertness, and circadian rhythm." Aviat Space Environ Med. 2004 Dec;75(12):1049-57. PMID: 15619859 , Dec-2004
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